Blessed G/Michael Catholic Secondary School

Semester 1 Mathematics Mid1 Examination for Grade10

Academic Year: 2024 / 2018 E.C. Time Allowed: 1:30'

Name: _____ **Grade & Sec:** Grade10 _____ **R.No:** _____

I. Choose the best answer among the given alternatives.

____ **1.** If f(x) = 2x^2 - 3x + 1, what is the value of f(-2)?
A. 3 **B.** 7 **C.** 15 **D.** 19

____ **2.** What is the solution set of the quadratic equation x^2 - 5x + 6 = 0?
A. {-2, -3} **B.** {2, 3} **C.** {-1, 6} **D.** {1, -6}

____ **3.** What is the vertex of the parabola given by y = (x - 4)^2 + 7?
A. (-4, 7) **B.** (4, 7) **C.** (4, -7) **D.** (-4, -7)

____ **4.** Which of the following relations is a function?
A. {(1, 2), (1, 3), (2, 4)} **B.** {(2, 5), (3, 7), (4, 9), (5, 11)}
C. {(3, 1), (3, 2), (3, 3)} **D.** {(0, 1), (0, -1), (2, 3)}

____ **5.** What is the domain of the function f(x) = sqrt(2x - 6)?
A. {x in R | x >= 3} **B.** {x in R | x > 3}
C. {x in R | x <= 3} **D.** {x in R | x != 3}

____ **6.** What is the remainder when P(x) = 2x^3 - 4x^2 + 5x - 3 is divided by (x - 2)?
A. 3 **B.** 5 **C.** 7 **D.** 11

____ **7.** If (x - 1) is a factor of P(x) = x^3 - 3x^2 + kx + 2, what is the value of k?
A. -2 **B.** 0 **C.** 1 **D.** 2

____ **8.** What is the vertex of the parabola f(x) = 2(x - 3)^2 + 5?
A. (-3, 5) **B.** (3, 5) **C.** (3, -5) **D.** (-3, -5)

____ **9.** What are the zeros of the polynomial f(x) = x^2 - 7x + 12?
A. x = -3 and x = -4 **B.** x = 3 and x = 4
C. x = 2 and x = 6 **D.** x = -2 and x = -6

____ **10.** The axis of symmetry of the quadratic function y = -x^2 + 6x - 8 is:
A. x = -3 **B.** x = 3 **C.** x = 6 **D.** x = 1

____ **11.** What is the solution of 3^(2x - 1) = 27?
A. x = 1 **B.** x = 2 **C.** x = 3 **D.** x = 4

____ **12.** Evaluate log_2(32) - log_3(81):
A. -1 **B.** 0 **C.** 1 **D.** 2

____ **13.** If f(x) = 3x - 2 and g(x) = x^2 + 1, find (f o g)(2):
A. 13 **B.** 15 **C.** 17 **D.** 21

____ **14.** What is the inverse of the function f(x) = (2x + 3) / 5?
A. f^(-1)(x) = (5x - 3) / 2 **B.** f^(-1)(x) = (5x + 3) / 2
C. f^(-1)(x) = (2x - 5) / 3 **D.** f^(-1)(x) = 5 / (2x + 3)

____ **15.** Which of the following is the discriminant of the equation 2x^2 - 4x + 3 = 0?
A. -8 **B.** 8 **C.** -40 **D.** 40

____ **16.** What is the degree and leading coefficient of the polynomial P(x) = -4x^5 + 7x^3 - 2x + 9?
A. Degree 5, leading coefficient -4 **B.** Degree 5, leading coefficient 9
C. Degree 3, leading coefficient 7 **D.** Degree 9, leading coefficient -4

____ **17.** If f(x) = x^2 - 4 and g(x) = x + 2, what is (f / g)(x) for x != -2?

A. x - 2 **B.** x + 2 **C.** x^2 - 2 **D.** 1

____ **18.** What is the minimum value of the quadratic function f(x) = x^2 - 6x + 13?

A. 4 **B.** 13 **C.** 3 **D.** -4

____ **19.** Solve the exponential equation: 2^(3x + 1) = 128

A. x = 2 **B.** x = 3 **C.** x = 7 **D.** x = 1

____ **20.** What is the value of log_5(1/125)?

A. -3 **B.** 3 **C.** 1/3 **D.** -1/3

____ **21.** Which of the following functions is strictly increasing on R?

A. f(x) = 2^x **B.** f(x) = (1/2)^x **C.** f(x) = -3x + 4 **D.** f(x) = -x^2

____ **22.** The solution set of the inequality x^2 - 5x + 6 < 0 is:

A. (2, 3) **B.** [2, 3]

C. (-infinity, 2) U (3, infinity) **D.** (-3, -2)

____ **23.** If log(x) + log(x - 3) = 1 (base 10), what is the value of x?

A. 5 **B.** -2 **C.** 2 **D.** 10

____ **24.** What is the sum of the roots of the quadratic equation 3x^2 - 12x + 5 = 0?

A. 4 **B.** -4 **C.** 5/3 **D.** -5/3

____ **25.** What is the range of the exponential function f(x) = 3^x - 2?

A. (-2, infinity) **B.** [-2, infinity)

C. (0, infinity) **D.** (-infinity, -2)